Project BaBaW (Bantay Baha Warning System)

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Problem Statement

Current flood warning systems have limitations in early prediction and usability. Many systems only issue alerts when water levels reach set thresholds, leaving limited time for residents to prepare or evacuate. Some also have interfaces that are difficult to understand during emergencies. These limitations can delay people's awareness of flood risks and their ability to respond effectively.

Features

Earlier Flood Awareness – Provides timely warnings based on current water levels, helping residents recognize potential flood risks sooner. Gives residents more time to prepare, secure their belongings, or evacuate to safer areas.

Faster Decision-Making – Presents water levels, rainfall, location, and warning status in a clear interface for easier understanding during emergencies.

Solution

Develop a simple Flood Warning System that monitors water levels and provides timely warnings based on current conditions. It will use a clear and user-friendly interface showing the location, water level, rainfall, and warning status, helping users quickly understand the flood situation and respond appropriately.

Gathering Parameters

The group will gather data (Water levels from the areas that belongs to the scope of the system) from a reliable source which is **Project Ligtas** (Live Flood Monitoring Philippines — Water Levels, Alerts & Flood Map | ProjectLIGTAS). On this website, a certain location of a Barangay has datasets that indicate the current water level status, the alert level for each water level, and a 24-Hour Water Level trend.

Prototype

The Prototype Setup consists of the following:

Arduino r3
Ultrasonic Sensors
Breadboard
Liquid Crystal Display (LCD)
Capacitors
Plywood
GSM SIM